Fiscal Year:	FY 2009	**Task Last Updated:**	FY 01/07/2009
PI Name:	Kronenberg, Amy Sc.D.		
Project Title:	Comparative Analysis of Charged Particle-Induced Autosomal Mutations in Murine Tissues and Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-8099	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	04/01/2007	**End Date:**	03/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Turker, Mitchell (Oregon Health & Science University)		
Grant/Contract No.:	NNJ07HC72I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Mutations are required in the transition from normal cells to cancer cells. Solid cancers in people arise primarily in epithelial cells. This proposal addresses the carcinogenic risk of exposure to energetic protons, the most abundant ion in galactic cosmic radiation (GCR) and solar particle events. The toxic effects of these exposures will also be measured. The studies outlined quantify mutation induction at a model, autosomal genetic locus (Aprt) located on mouse chromosome 8. C57BL/6 x DBA2 F1 Aprt heterozygous mice or kidney epithelial cells from this mouse strain will be irradiated to determine dose responses for cell killing and Aprt mutation in kidney epithelial cells exposed in the mice, or in kidney epithelial cells irradiated in culture. This will allow assessment of the role of the tissue microenvironment on modulation of mutation risk and on persistence of radiation-induced mutants in the kidney. Molecular analysis will determine the mechanisms leading to proton-induced Aprt mutants in exposed kidneys or in kidney cells irradiated in culture. It is predicted that proton exposure will result in mutants that lose the Aprt gene and other genes that are located		

	in nearby regions of mouse chromosome 8. Studies are proposed to test if protracted exposure lessens the frequency of proton-induced mutants while also lowering the risk of killing kidney epithelial cells. The toxic and mutagenic effects of sequential exposure to protons and iron ions will be assessed (with iron ions serving as a representative heavy ion in the GCR) to test whether the risks are additive. Results from these studies will help reduce uncertainties in estimating the risk of cancer for astronauts exposed to protons alone or in mixed radiation fields. They will also provide detailed knowledge of the types of mutations that are created by these exposures that can contribute to the development of cancer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will have a direct impact on our understanding of proton-induced toxicity and mutations in a mammalian epithelial tissue. This has relevance to cancer therapy involving proton exposures that not only target malignant tissue, but which also involve limited exposure of normal tissues that can develop late complications as a result of proton exposure. In addition, our in vivo/in vitro approach also illuminates the importance of the normal tissue environment on the modulation of radiation sequelae.
Task Progress:	This project is designed to evaluate the genotoxicity of energetic protons using mouse kidney epithelium as an indicator. Studies are proposed to evaluate the influence of the tissue microenvironment on cytotoxicity and mutation induction at a model autosomal locus, Aprt, located on chromosome 8. Other studies are planned that will evaluate the influence of fractionated exposures to high energy protons and to evaluate combined effects of exposure to protons followed by densely ionizing Fe ions. During this year, studies continued with an analysis of in situ cytotoxicity from the first experimental runs at NSRL in 2007. The results indicate minimal toxicity to kidney epithelial cells in situ for exposures of up to 4 Gy of sparsely ionizing 1 GeV protons monitored at the early harvest time. Similar results were obtained for assays performed at the late harvest time. A second experimental run was performed in Spring 2008, with an expansion of the dose range to 5 Gy. Results from the early harvest time in the second run suggest that expansion of the dose range is accompanied by reduced cloning efficiencies for kidney epithelial cells. Analysis of Aprt mutant frequencies for the in situ exposures is ongoing, with evidence for induction of autosomal mutations at this intermediate point in the study. Large stocks of kidney epithelial cells derived from primary clones were expanded for in vitro studies and have provided data for the first series of proton exposures (doses up to 5 Gy) demonstrating a mutagenic effect of sparsely ionizing, high energy protons at short times post-irradiation. Additional cultures are in preparation for replicate experiments at the NSRL during 2009.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Comparative analysis of charged particle-induced mutations in murine cells and tissues." NASA Human Research Program Investigators' Workshop, Houston, TX , February 2008. NASA Human Research Program Investigators' Workshop, Houston, TX , February 2008. , Feb-2008
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Autosomal mutagenesis after charged particle exposure of mouse kidney epithelial cells." NASA Space Radiation Health Investigators' Workshop, Philadelphia, PA, June 2008. NASA Space Radiation Health Investigators' Workshop, Philadelphia, PA, June 2008. , Jun-2008
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Comparative analysis of charged particle-induced autosomal mutations in murine cells and tissues." 27th COSPAR Scientific Assembly, Montreal, Canada, July 2008. 27th COSPAR Scientific Assembly, Montreal, Canada, July 2008. , Jul-2008
Abstracts for Journals and Proceedings	Kronenberg A, Gauny S, Kwoh E, Dan C, Turker M. "Charged particle-induced autosomal mutations and the influence of the tissue microenvironment." 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 2008. 54th Annual Meeting of the Radiation Research Society, Boston, MA, September 2008. , Sep-2008